

Jimmy Shong

650-681-7291 | jimmysh341@gmail.com | linkedin.com/in/jimmy-shong | github.com/Jiminitor | [Portfolio Website](#)

EDUCATION

University of Illinois Urbana-Champaign

Masters of Science in Computer Science; Advisor: Prof. Fan Lai

Champaign, IL

Aug. 2024 – May 2026

New York University

Bachelor of Science in Computer Science, Minor in Mathematics and Cybersecurity

New York, NY

Aug. 2020 – May 2024

EXPERIENCE

ML Engineer Intern

Waymo

May 2025 – Aug 2025

Mountain View, CA

- Identified and resolved performance bottlenecks by diving deep into production JAX planner-model code, using xprof to profile end-to-end latency, implementing targeted optimizations and bug fixes.
- Led the ML Infra org's initiative to quantize planner models, performing statistical analysis, building evaluation pipelines, assisted in key downstream technical decisions, and delivering end-to-end quantization implementations.
- Coordinated across five different teams to integrate and ship quantized versions of three separate planner models

Student Researcher

University of Illinois Urbana-Champaign, FANS Lab

Aug. 2024 – Present

Champaign, IL

- Lead the research effort on **COMET** to efficiently find near-optimal configurations for cloud ML jobs

Research Intern

Massachusetts Institute of Technology, MIT HAN Lab

May 2023 – May 2024

Cambridge, MA

- Implemented matrix multiplication optimizations for TinyChatEngine, a high-performance LLM inference library
- Built **TinyVoiceChat**, a tool that allows users to interface with a quantized LLM with their voice on edge devices
- Researched CoreML and Metal backends to improve TinyChatEngine's performance on Apple Silicon hardware

Unsupervised ML Research Lead

New York University, AI for Scientific Research

Sep. 2021 – May 2022

New York, NY

- Engineered a Python library that analyzed macrophage trajectories and predicted their diffusion states
- Presented our work to researchers at the University of Groningen who utilized our tool in their experiment
- Performed data visualization and analysis on the sleep-wake dynamics of mice for scientists at NYU Abu Dhabi

PUBLICATIONS

Comet: Cost-Effective Cloud ML Execution with Mixed GPUs

J. Shong, Y. Ding, Y. Jiang, H. Chen, L. Jing, G. Zhang, A. Akella, F. Lai

- In submission to **SIGCOMM 2026**
- We propose a novel planner to find near-optimal cloud ML configs, overcoming the complexity of heterogeneous hardware and the ever-evolving frameworks, achieving up to **2.72× throughput gain** over expert-tuned settings.

PROJECTS

Improved Model Parallelism for Distributed LLM Training | *Python*

Aug. 2024 – Dec 2024

- Identified inefficiencies in Metis, a state-of-the-art auto-parallelizer for heterogeneous clusters
- Implemented a novel search strategy, achieving up to 2× speedup without compromising model accuracy
- Authored a draft research paper detailing the inefficiencies, proposed solution, and experimental results

Exploring SFT Methods for LLMs | *Pytorch, SLURM*

Apr. 2024 – May 2024

- Instruct-tuned Llama3-8B using parameter-efficient finetuning (PEFT) methods with LlamaFactory
- Evaluated the training speed, memory usage, and model performance of these SFT methods

Google Suite Task Manager | *Flask, MongoDB, Ruff, mypy, Python PDM, CircleCI*

Jan. 2024 – May 2024

- Developed a task manager that aggregates all tasks created in both Google Tasks and other Google Suite products
- Implemented fundamental task management features currently available in Google Tasks

NetArmor | *Python, TurboGears, FastAPI, SQLAlchemy*

Sep. 2023 – Dec. 2023

- Developed a tool for website owners to scan their sites for vulnerabilities and contact cybersecurity professionals
- Engineered a PostgreSQL database using SQLAlchemy to store all necessary data

TEACHING EXPERIENCE

Teaching Assistant

Aug. 2024 – Present

University of Illinois Urbana-Champaign

Champaign, IL

- Lectured on modern transformers and FlashAttention in front of 90+ students taking CS 598: Systems for GenAI and CS 498: ML Systems
- Hosted lab sections and office hours for students in CS 598: Systems for GenAI, CS 498: ML Systems, and STAT-107: Data Science Discovery

Computer Science Tutor

Jan 2023 – May 2024

NYU Polytechnic Tutoring Center

New York, NY

- Answered questions from NYU CS students taking the required computer science courses
- Recorded explanation videos for answer keys of mock exams developed by the PTC

Instructor

June 2022 – Jul 2022

BlueStamp Engineering

Palo Alto, CA

- Taught practices and principles of engineering to a class of twenty high school students
- Led projects such as an intelligent door lock, a smart mirror, and Alexa home automation

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL (MySQL), JavaScript, HTML/CSS

Frameworks: PyTorch, JAX, TensorFlow, CUDA, CoreML, Metal, MongoDB, FastAPI, Flask, TurboGears

Developer Tools: Git, Docker, CircleCI, Ruff, mypy, Python PDM, AWS, SLURM, Linux (Ubuntu, Kali), MacOS

Libraries: Megatron-LM, vLLM, NumPy, pandas, Hugging Face, Matplotlib, SQLAlchemy